
Hitting Time Guarantees for Blind Ballistic “Billiard” Search

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Problem

Let the state space be a 2-dimensional discrete torus $S = \mathbb{Z}_N \times \mathbb{Z}_M$. The total size of the state space is $|S| = NM$, with N, M hidden from the agent. The agent, blind to its starting position $\tilde{x}$, is tasked with finding an arbitrary target node $x^* \in S$. All bounds below hold uniformly over $\tilde{x}$ and x^* .

Assume without loss of generality that $N \leq M$. We enforce a bounded aspect ratio assumption, meaning there exists a constant $R \geq 1$ such that $M \leq R \cdot N$. Let the maximum dimension be $D = M$. Since $N \leq M \leq RN$, we have $|S| \leq D^2 \leq R \cdot |S|$, meaning $D = \Theta(\sqrt{|S|})$. The lower bound $D^2 \geq |S|$ holds for every torus; the aspect-ratio assumption is what provides the upper bound.

The Iterative Ballistic Process $\mathcal{B}$

Since D is not known in advance, sampling uniformly up to D is impossible. The search instead scales dynamically, operating in progressively larger *phases*. Fix a constant $c > 0$, a parameter of the algorithm that does not depend on N or M .

Initialize a phase counter $j = 1$. During Phase j , the agent executes the following routine:

- (i) The agent guesses a maximum dimension $D_j = 2^j$.
- (ii) The agent runs a continuous sequence of “epochs” (linear streaks). In each epoch:
 - (1) The agent selects a direction vector v uniformly at random from the set of axial and diagonal generators: $V = \{(1, 0), (0, 1), (1, 1), (1, -1)\}$.
 - (2) The agent selects a streak length L uniformly at random from the integer interval $[1, D_j]$.
 - (3) The agent traverses the grid by taking L consecutive 1-cell steps along v , checking every node along the path for the target state x^* .
- (iii) The agent continues initiating epochs until the cumulative number of single-cell steps taken in Phase j reaches or exceeds $c \cdot D_j^2$. The agent completes its final epoch rather than truncating it mid-streak, so that every epoch is a complete, independent draw from the same streak distribution. Because the maximum length of an epoch is D_j , the total step budget overshoots by less than D_j , so Phase j costs at most $cD_j^2 + D_j = \Theta(D_j^2)$ steps. Conversely, while fewer than cD_j epochs have been run, fewer than cD_j^2 steps have been taken; hence every phase deterministically runs at least $\lceil cD_j \rceil$ epochs.
- (iv) If the target x^* is not found by the end of the step budget, the agent increments j to $j + 1$ and begins the next phase from its current location.

We seek to bound $\mathbb{E}[C_{x^*}]$, the expected total number of single-cell steps taken across all phases before the target x^* is visited.

“Activation Phase” and Spectral Gap

In early phases where $D_j \ll D$, the agent’s streaks are too short to efficiently mix across the longest dimension of the torus. However, because D_j doubles every phase, there will

be a specific phase j^* , which we call the *Activation Phase*, that is the first phase to satisfy $D_{j^*} \geq D$. By definition of the doubling sequence, $D \leq D_{j^*} < 2D$ whenever $D \geq 2$ (if $D = 1$, then $|S| = 1$ and there is nothing to prove). Every later phase $j \geq j^*$ also satisfies $D_j \geq D$, and the analysis below applies to all of them simultaneously.

Fix a phase $j \geq j^*$. The sequence of epoch *start points* $X_0, X_1, \dots$ forms a discrete-time Markov chain on the abelian group S , where each transition corresponds to a full epoch trajectory: $X_{t+1} = X_t + L_t v_t$ with (v_t, L_t) i.i.d. Writing μ_j for the law of the displacement Lv , the transition matrix is $P_j(x, y) = \mu_j(y - x)$. The transition kernel is translation-invariant, so P_j is a group convolution operator. Convolution operators on an abelian group commute with one another, and the adjoint P_j^* is convolution with the reflected law, so P_j is normal ($P_j P_j^* = P_j^* P_j$). While not necessarily symmetric, P_j is doubly stochastic, so the uniform distribution $\pi(x) = 1/|S|$ is stationary; it is the unique stationary measure because the chain is irreducible (taking $L = 1$ shows that μ_j charges $(1, 0)$ and $(0, 1)$, which generate S). Normality is what we actually use: P_j is diagonalized by an orthonormal basis, so the moduli of its non-trivial eigenvalues control its mixing exactly as they would for a reversible chain.

The eigenvalues of this kernel correspond to the Fourier characters of $\mathbb{Z}_N \times \mathbb{Z}_M$, namely

$$\chi_w(z) = e^{2\pi i(w_1 z_1/N + w_2 z_2/M)}$$

for frequency vectors $w = (w_1, w_2) \in S$: we have $P_j \chi_w = \lambda_w \chi_w$ with

$$\lambda_w = \frac{1}{4} \sum_{v \in V} \left(\frac{1}{D_j} \sum_{\ell=1}^{D_j} \exp \left(2\pi i \ell \left(\frac{v_1 w_1}{N} + \frac{v_2 w_2}{M} \right) \right) \right),$$

and $w = (0, 0)$ gives the trivial eigenvalue $\lambda_{(0,0)} = 1$.

Let

$$\theta_v(w) := \frac{v_1 w_1}{N} + \frac{v_2 w_2}{M},$$

and write $\|\theta\|$ for the distance from θ to the nearest integer. For any non-zero frequency w , already one of the two *axial* generators produces a phase bounded away from 0 on the scale $1/M$ (the diagonal generators are not needed for this step): if $w_1 \neq 0$, then $\|\theta_{(1,0)}(w)\| = \|w_1/N\| \geq 1/N \geq 1/M$; otherwise $w_2 \neq 0$ and $\|\theta_{(0,1)}(w)\| = \|w_2/M\| \geq 1/M$. For this non-aligned vector, with $\theta = \theta_v(w)$, the magnitude of the inner geometric sum satisfies

$$\left| \frac{1}{D_j} \sum_{\ell=1}^{D_j} e^{2\pi i \ell \theta} \right| = \frac{1}{D_j} \left| \frac{\sin(\pi D_j \theta)}{\sin(\pi \theta)} \right| \leq \frac{1}{2D_j \|\theta\|} \leq \frac{M}{2D_j} \leq \frac{1}{2},$$

where we used $|\sin(\pi D_j \theta)| \leq 1$, Jordan’s inequality $|\sin(\pi \theta)| = \sin(\pi \|\theta\|) \geq 2\|\theta\|$, and $D_j \geq D = M$. In other words, because the minimum non-zero phase is at least $1/M = 1/D$ and $D_j \geq D$, the summation covers enough of the group to destructively interfere, and its magnitude is bounded above by $1 - \delta$ with the explicit constant $\delta = 1/2$, which does not even depend on R .

Bounding the other three directions trivially by 1, the non-trivial eigenvalues are bounded strictly away from the unit circle:

$$\lambda_\star := \max_{w \neq (0,0)} |\lambda_w| \leq \frac{1}{4} \left(3 + \frac{1}{2} \right) = 1 - \frac{\delta}{4} = \frac{7}{8} \quad \text{for every phase } j \geq j^*.$$

Since the normalized characters $\chi_w / \sqrt{|S|}$ form an orthonormal eigenbasis of P_j , this constant spectral gap gives, for every $f : S \rightarrow \mathbb{C}$ with $\sum_x f(x) = 0$,

$$\|P_j^t f\|_2 \leq \left(\frac{7}{8}\right)^t \|f\|_2 \quad (t \geq 0), \quad \text{where } \|f\|_2^2 := \sum_{x \in S} |f(x)|^2, \quad (1)$$

i.e., an effective relaxation time of $\mathcal{O}(1)$ (at most 8) epochs.

Stationary Hitting Probability

During Phase j^* , the expected length of an epoch is

$$\mathbb{E}[L] = \frac{D_{j^*} + 1}{2}.$$

Because $D_{j^*} < 2D = 2M$, a streak along the longest dimension (or along a diagonal, whose orbit has $\text{lcm}(N, M) \geq M$ nodes) wraps at most once. However, along the shorter dimension N , it may wrap up to $\lfloor 2M/N \rfloor \leq 2R$ times. Since the aspect ratio R is bounded by a constant, the number of distinct nodes explored in an epoch is reduced by at most a factor of $2R$. Thus, the number of distinct nodes remains proportional to the streak length.

If a trajectory begins from a start point drawn from the stationary uniform distribution π , the stationary probability $P_\pi(H)$ that this trajectory intersects x^* is bounded below by accounting for the maximum overlap:

$$P_\pi(H) \geq \frac{1}{2R} \sum_{\ell=1}^{D_{j^*}} P(L = \ell) \frac{\ell}{|S|} = \frac{D_{j^*} + 1}{4R|S|} = \Theta\left(\frac{D_{j^*}}{|S|}\right).$$

Heuristically, then, about $1/P_\pi(H) = \mathcal{O}(|S|/D_{j^*}) = \mathcal{O}(D_{j^*})$ epochs should suffice to hit x^* (using $|S| \leq D^2 \leq D_{j^*}^2$), which matches the $\Theta(D_{j^*})$ epochs allocated to Phase j^* . Turning this heuristic into a proof requires care, for three reasons.

- (a) There is no general bound of the form $\mathbb{E}_\pi[T_{x^*}] = \mathcal{O}\left(\frac{1}{1 - \max|\lambda|} + \frac{1}{P_\pi(H)}\right)$ for normal chains on vertex-transitive graphs. For example, for the lazy simple random walk on the $n \times n$ torus (reversible, hence normal), the relaxation time is $\Theta(n^2)$ and $1/\pi(x^*) = n^2$, yet $\mathbb{E}_\pi[T_{x^*}] = \Theta(n^2 \log n)$.
- (b) The event H concerns the entire streak of an epoch, not the start-point chain X_t entering a set, so standard hitting-time results for X_t do not apply directly.
- (c) Phase j^* begins wherever Phase $j^* - 1$ ended, not from stationarity; and matching the allocated epochs to an expected hitting time only up to unspecified constants does not, by Markov’s inequality, yield a success probability bounded away from zero.

We therefore prove a worst-case statement directly, using the spectral gap (1) and a second-moment argument.

A Worst-Case Block Hitting Lemma

Fix a phase $j \geq j^*$ and view its epochs as an i.i.d. sequence. To keep the second moment under control, we only count hits made by *vertical* streaks. Let G_t be the event that epoch t has $v_t = (0, 1)$ and its streak passes through x^* , and let $g_j(y) := P(G_t \mid X_t = y)$. For y in the column of x^* (i.e., $y_1 = x_1^*$), let $d(y) \in \{1, \dots, M\}$ be the number of $(0, 1)$ -steps needed to go from y to x^* (with $d(x^*) = M$). Then

$$g_j(y) = \frac{1}{4} P(L \geq d(y)) \quad \text{if } y_1 = x_1^*, \quad g_j(y) = 0 \quad \text{otherwise.}$$

We need three elementary estimates.

- (E1) *Mean.* Since L is uniform on $\{1, \dots, D_j\}$ with $D_j \geq M$, we have $\sum_{d=1}^M P(L \geq d) = \mathbb{E}[\min(L, M)] \geq \frac{M+1}{2}$. Hence

$$p_j := \pi(g_j) = \frac{1}{|S|} \sum_{y \in S} g_j(y) \geq \frac{M+1}{8NM} > \frac{1}{8N}, \quad \text{and also} \quad p_j \leq \frac{M}{4|S|} \leq \frac{1}{4}.$$

- (E2) *Size.* Since $0 \leq g_j \leq \frac{1}{4}$, we have $\|g_j\|_2^2 \leq \frac{1}{4} \sum_y g_j(y) \leq \frac{M}{16}$.

- (E3) *Spread of a single epoch.* For $z \in S$ and $v \in V$, the solutions ℓ of $\ell v = z$ form a single residue class modulo the order $o(v)$ of v in S , so at most $D_j/o(v) + 1$ of them lie in $\{1, \dots, D_j\}$. Since $o((1, 0)) = N$, $o((0, 1)) = M$, $o((1, \pm 1)) = \text{lcm}(N, M) \geq M$, and $D_j \geq M \geq N$,

$$\max_{z \in S} \mu_j(z) \leq \frac{1}{4} \left(\frac{1}{N} + \frac{3}{M} \right) + \frac{1}{D_j} \leq \frac{2}{N}.$$

Decorrelation. For $t \geq 1$ and $y \in S$, write $P_j^t g_j(y) - p_j = \sum_z \mu_j(z) f(y + z)$ with $f := P_j^{t-1}(g_j - p_j)$. By Cauchy–Schwarz, (1), (E2), and (E3), using $\|\mu_j\|_2^2 \leq \max_z \mu_j(z)$ and $\|g_j - p_j\|_2 \leq \|g_j\|_2$,

$$|P_j^t g_j(y) - p_j| \leq \|\mu_j\|_2 \|f\|_2 \leq \sqrt{\frac{2}{N}} \left(\frac{7}{8}\right)^{t-1} \sqrt{\frac{M}{16}} \leq \varepsilon_t := \left(\frac{7}{8}\right)^{t-1} \sqrt{\frac{R}{8}}, \quad (2)$$

and $\sum_{t \geq 1} \varepsilon_t = 2\sqrt{2R} =: a$.

Lemma. Let $\gamma := 34\sqrt{2R}$ and $T_* := \lceil \gamma D_{j^*} \rceil$. In every phase $j \geq j^*$, for any epoch index s and conditionally on everything that happened before epoch s , the probability that at least one of the epochs $s, \dots, s + T_* - 1$ visits x^* is at least $1/7$.

Proof. Because the epochs are i.i.d., by the Markov property it suffices to consider epochs $0, \dots, T - 1$ started from an arbitrary point $X_0 = x$, where $T := 1 + \lceil 2a/p_j \rceil$. By (E1), $T < 2 + 16aN < 17aN = \gamma N \leq \gamma D_{j^*} \leq T_*$, so these epochs lie within the block. Let $Z := \sum_{t=0}^{T-1} \mathbf{1}_{G_t}$.

First moment. Since $P(G_t) = P_j^t g_j(x)$, (2) gives $\mathbb{E}[Z] \geq \sum_{t=1}^{T-1} (p_j - \varepsilon_t) \geq (T-1)p_j - a \geq a$.

Second moment. The event G_s is determined by the first $s+1$ epochs, and later epochs are fresh, so for $s < t$ we have $P(G_t \mid \text{epochs } 0, \dots, s) = P_j^{t-s-1} g_j(X_{s+1})$. Using $g_j \leq \frac{1}{4}$ when $t = s+1$ and (2) otherwise,

$$\sum_{t=s+1}^{T-1} P(G_s \cap G_t) \leq P(G_s) \left(\frac{1}{4} + \sum_{m=1}^T (p_j + \varepsilon_m) \right) \leq P(G_s) \left(\frac{1}{4} + Tp_j + a \right).$$

Summing over s and using $Tp_j \leq 2a + 2p_j \leq 2a + \frac{1}{2}$,

$$\mathbb{E}[Z^2] = \mathbb{E}[Z] + 2 \sum_{s < t} P(G_s \cap G_t) \leq \mathbb{E}[Z] \left(\frac{3}{2} + 2Tp_j + 2a \right) \leq \mathbb{E}[Z] \left(\frac{5}{2} + 6a \right).$$

By the second-moment (Cauchy–Schwarz) inequality,

$$P(Z \geq 1) \geq \frac{\mathbb{E}[Z]^2}{\mathbb{E}[Z^2]} \geq \frac{\mathbb{E}[Z]}{\frac{5}{2} + 6a} \geq \frac{a}{\frac{5}{2} + 6a} \geq \frac{1}{7},$$

where the last step uses $a = 2\sqrt{2R} \geq 2\sqrt{2} > \frac{5}{2}$. $\square$

In particular, from *any* start, the expected number of epochs until x^* is visited is at most $7T_* = \mathcal{O}(D_{j^*}) = \mathcal{O}(|S|/D_{j^*})$ (the last step because $D_{j^*}^2 < 4D^2 \leq 4R|S|$), which is the worst-case version of the heuristic above. If moreover $c \geq \gamma + 1$, then Phase j^* contains a full block, so the activation phase itself succeeds with probability at least $1/7$. For smaller c this may take a few more phases, which only affects constants.

Bounding the Expected Total Steps

If the agent fails to find the target in Phase j^* , it proceeds to Phase $j^* + 1$, where the allocated steps quadruple. Crucially, the hitting probability does not remain constant; the failure probability decays doubly exponentially in k .

In Phase $j^* + k$, the allocated maximum dimension is $D_{j^*+k} = 2^k D_{j^*}$, and the step budget is $\Theta(4^k D_{j^*}^2)$. As the epoch lengths grow past the maximum dimension of the torus, the hitting probability per epoch saturates rather than grows: once $L \geq M$, a vertical streak covers a full column (a horizontal streak covers a full row of N nodes, and a diagonal streak a full diagonal orbit of $\text{lcm}(N, M)$ nodes). By (E1) the vertical-hit probability never drops below $1/(8N)$, and the lemma applies to every phase $j \geq j^*$ with the *same* block length T_* . Meanwhile, Phase $j^* + k$ deterministically runs at least $\lceil c 2^k D_{j^*} \rceil$ epochs, which contain at least

$$b_k := \left\lfloor \frac{c 2^k}{\gamma + 1} \right\rfloor$$

disjoint blocks of T_* consecutive epochs (using $T_* \leq \gamma D_{j^*} + 1 \leq (\gamma + 1)D_{j^*}$). Applying the lemma block by block, the probability q_k that Phase $j^* + k$ fails, conditionally on any history before it, satisfies

$$q_k \leq \left(\frac{6}{7}\right)^{b_k} \leq \frac{7}{6} \left(\frac{6}{7}\right)^{c 2^k / (\gamma + 1)} = \frac{7}{6} \exp(-\beta 2^k), \quad \beta := \frac{c \ln(7/6)}{\gamma + 1} > 0.$$

In particular, reaching Phase $j^* + k$ requires failing Phase $j^* + k - 1$, which happens with probability at most q_{k-1} .

The total number of single-cell steps taken up to the end of Phase j^* is dominated by a geometric series:

$$\sum_{i=1}^{j^*} \Theta(D_i^2) = \sum_{i=1}^{j^*} \Theta(4^i) = \Theta(4^{j^*}) = \Theta(D_{j^*}^2).$$

The expected total step cost is at most the maximum number of steps taken up to Phase j^* plus, for each $k \geq 1$, the (deterministically bounded) cost of Phase $j^* + k$ times the probability of reaching it:

$$\mathbb{E}[C_{x^*}] \leq \mathcal{O}(D_{j^*}^2) + \sum_{k=1}^{\infty} q_{k-1} \cdot \mathcal{O}(4^{j^*+k}).$$

Substituting the bound on q_{k-1} :

$$\mathbb{E}[C_{x^*}] \leq \mathcal{O}(D_{j^*}^2) + \sum_{k=1}^{\infty} \frac{7}{6} \exp(-\beta 2^{k-1}) \cdot \mathcal{O}(4^{j^*+k}).$$

Because the doubly exponential decay of $\exp(-\beta 2^{k-1})$ overpowers the geometric growth of 4^k (the ratio of consecutive terms is $4e^{-\beta 2^{k-1}} \rightarrow 0$), the series converges to a constant multiple, depending only on c and R , of $4^{j^*} = D_{j^*}^2$. Finally, $D_{j^*} < 2D$ and the bounded aspect ratio $D^2 \leq R|S|$ give

$$\mathbb{E}[C_{x^*}] = \mathcal{O}(D_{j^*}^2) = \mathcal{O}(D^2) = \mathcal{O}(|S|).$$

This concludes the proof. The blind agent achieves linear expected hitting time, uniformly over its starting position and the target, adapting to the unknown dimensions of the torus solely through the doubling schedule $D_j = 2^j$.